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Role of basic chemistry education in clinical pharmacy requirements: evidence from cross-sectional study and narrative review

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Abstract

Background Basic chemistry knowledge is important for the work requirements of clinical pharmacists. However, it is not clear which types of basic chemistry teaching are the most important. The purpose of this study was to determine the most important types of chemistry education for the professional requirements of clinical pharmacists.

Methods Section I, a cross-sectional study was conducted through a questionnaire survey administered from December 2024 to February 2025 to collect data on the importance of analytical chemistry, inorganic chemistry, organic chemistry, medicinal chemistry, experimental medicinal chemistry, physical chemistry and biochemistry among undergraduate and graduate students who had received clinical pharmacist training, current clinical pharmacists or other personnel. Second II, we conducted a narrative review up through March 2025 that included our cross-sectional study and other publications to highlight the importance of various chemical education methods.

Results In Section I, a total of 600 participants were included in the cross-sectional study of the questionnaire survey. We found that all categories of participants believed that there was no disconnect between chemistry education and clinical rotation and that knowledge of medicinal chemistry and biochemistry were considered more important among participants with different levels and backgrounds. In addition, in terms of the application of chemical knowledge, we found that undergraduate and graduate students tended to infer the solubility and adverse reactions of medications through the chemical structural formulas in the instructions. In Section II, the narrative review revealed that medicinal chemistry and biochemistry play particularly important roles in meeting the demands of clinical pharmacists.

Conclusion Through cross-sectional research and narrative review, this study revealed that medicinal chemistry and biochemistry education are more important for the professional requirements of clinical pharmacists, especially for junior clinical pharmacists.

Keywords Chemistry education, Medicinal chemistry, Biochemistry, Clinical pharmacy, Cross-sectional study

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Introduction

Medications are chemical substances that affect the physiological functions of the body; they are composed of a single or composite chemical substance, and their active ingredients exert their effects through chemical mechanisms. Even for medicines of natural origin, their active ingredients are still specific chemical molecules (such as alkaloids, glycosides, etc.) that ultimately regulate physiological functions through chemical pathways [1–3]. Clinical pharmacists need to understand the chemical structures and interactions of medications in their daily work. Therefore, the demand for chemical knowledge is crucial for the professional level of clinical pharmacists, and basic knowledge training for clinical pharmacists begins at the undergraduate level and generally includes the study of eight major chemical disciplines, namely, analytical chemistry, inorganic chemistry, organic chemistry, medicinal chemistry, medicinal chemistry experiment, physical chemistry, biochemistry and natural medicinal chemistry [4–6].

Clinical pharmacists serve as a critical interdisciplinary bridge between pharmacy and medicine [7]. They shoulder the weighty responsibility of improving the quality of medication by ensuring the safety, rationality, and effectiveness of drugs for patients and undertake the important mission of combating the tenacious struggle between humans and diseases. The importance of basic chemistry courses is self-evident; for example, knowledge of medicinal chemistry helps clinical pharmacists understand the chemical structures, physicochemical properties, and in vivo metabolism of medications. Moreover, knowledge of medicinal chemistry can help clinical pharmacists conduct research on drug interactions, thereby avoiding potential adverse drug reactions. Additionally, medicinal chemistry knowledge can be used by clinical pharmacists in the development and evaluation of new drugs, thereby promoting the development of the pharmaceutical industry [8–10]. Knowledge of biochemistry can help clinical pharmacists further improve their attitudes towards collaboration with physicians and help them understand the roles of vitamins and cofactors in metabolism. In addition, biochemical knowledge enables clinical pharmacists to clarify the structure of DNA/RNA and its metabolic processes, as well as to further understand the mechanisms of action of antibacterial and anticancer agents that affect these processes. Furthermore, such knowledge can enable further research on pharmacogenomics [11–13].

Clinical pharmacy education in China started relatively late, and the education system is still in the stage of continuous experimentation and exploration and is not yet fully mature [14, 15]. There are also many matters, such as the rigid connection between various chemistry courses as well as obvious module segmentation between chemistry, pharmacy, and medical courses, which fail to truly integrate into one course and form professional characteristics. It is also unclear which basic chemistry course is most important for meeting the routine requirements of clinical pharmacists. Therefore, the aim of this study was to identify the basic chemistry courses that have the greatest value for the work of clinical pharmacists, with the aim of increasing their teaching hours and further integrating them with medication and medical courses to form a basic chemistry course system with clinical pharmacy professional characteristics that can be applied to aid in the practical work of clinical pharmacists.

Methods

Overall framework

Our research framework is based on an explanatory sequential design. Section I, we utilized a cross-sectional study from questionnaire surveys to determine the most valuable forms of chemistry education based on the professional needs of clinical pharmacists. Second II, a narrative review of this cross-sectional study and relevant publications was performed to support and expand the findings of the cross-sectional study to determine the most important chemistry education courses for the professional needs of clinical pharmacists.

Section I: Cross-sectional study

Participants

The participants we included were senior undergraduate students in clinical pharmacy (fourth and fifth year) and senior graduate students in clinical pharmacy (second and third year) who had received clinical pharmacist training and had undergone clinical rotations in at least two departments for more than six months. Therefore, all of these students could be considered junior clinical pharmacists. Current clinical pharmacists or personnel who had received clinical pharmacist training, as well as those who had been exposed to clinical training and are currently engaged in other work, were also included.

In Chinese hospitals, to become a clinical pharmacist, a person must be an undergraduate student majoring in pharmacy or clinical pharmacy and pass the exam. During the undergraduate stage, the students must have studied eight basic chemistry courses, including analytical chemistry, inorganic chemistry, organic chemistry, medicinal chemistry, experimental medicinal chemistry, physical chemistry, biochemistry and natural medicinal chemistry, as mentioned in the questionnaire. In addition, all pharmacy majors in China must have a national professional teaching certification to ensure that respondents have a common baseline understanding of the course content. The survey was conducted from December 2024 to March 2025, and most participants were

from a pharmaceutical university in Northeast China, with no age restriction.

Study design and questionnaire

This was a cross-sectional study, and participants were asked to complete an online survey after providing consent. This questionnaire was designed to be anonymous by our research group (Table S1) and included a demographic questionnaire and questions designed on the basis of the application and proportion of chemistry in clinical rotation. The demographic questionnaire was quantitative in nature and included questions about participant type, age, and gender and queried participants on whether they felt there was a disconnect between theory and practice in basic chemistry education and what they thought regarding the weight of basic chemistry education. In addition, the participants' perceptions of the importance of the eight chemical subjects of analytical chemistry, inorganic chemistry, organic chemistry, medicinal chemistry, physical chemistry, biochemistry and experimental medicinal chemistry were assessed. Participants rated the importance on a three-point scale: very important, generally important, and unimportant. This three-point scale allows participants to answer as quickly as possible, which is feasible and easy to obtain genuine feelings. Moreover, some questions related to the application of chemical knowledge in clinical pharmacist work, such as have you read the structure of the instruction manual, have you considered the metabolic properties of the drug on the basis of its structural formula, have you considered drug solubility or lipid solubility on the basis of the structural formula, have you considered drug side effects on the basis of the structural formula and have you ever felt that knowledge of medicinal chemistry is very important in your clinical rotation as a clinical pharmacist. The above questions all produced categorical results, and their settings are consistent with some other survey-based publications [16, 17].

Statistical analysis of survey results

Categorical results are expressed as numbers and proportions, and to explore which factor was the most important, we treated the numbers and percentages as dichotomous variables. Factors selected as "important" by over 70% of the participants were considered to have a great impact on the application of clinical pharmacists. Comparisons of intergroup importance differences among different participant categories were performed using a chi-square test. Multivariate logistic regression was performed to exclude the influence of demographic factors such as gender and age, and the control group was set as the other participant group to further evaluate the intergroup differences. A p-value < 0.05 was considered to indicate statistical significance. All analyses were performed with the statistical analysis software SPSS 24.0.

Second II: Narrative review

Data sources, extraction and quality assessment

The narrative review section of this study follows the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) checklist [18]. Electronic searches were performed from inception to March 2025 in Pubmed, the Cochrane Library, Web of Science, Sinomed, China National Knowledge Infrastructure (CNKI) and Wanfang for eligible articles. The screening was performed by two independent, experienced reviewers (ZYS and XZA) with the keywords "importance of chemistry education" without language limitation (see details in Table S2). Cross-sectional studies with data on the importance of one or more basic chemistry education programs for clinical pharmacist requirements were also included, as well as the cross-sectional study we had just conducted. Moreover, the American Agency for Healthcare Research and Quality (AHRQ) scale was used to evaluate the quality of the included studies. The AHRQ included 11 items with answers of "yes", "no, " or "unclear". If the answer is "no" or "unclear", the score is 0, and if the answer is "yes", a score of 1–3 indicates low quality, 4–7 indicates medium quality and 8–11 indicates high quality [19]. The above steps were completed independently by two reviewers.

Data synthesis

For the included studies, we conducted a narrative review and our cross-sectional study data. We used quantitative data for the importance of each basic chemistry course and compared the importance data using odds ratios (ORs) with 95% confidence intervals (95% CIs). We ranked each type of chemistry using a surface under the cumulative ranking curve (SUCRA), and a high SUCRA percentage indicated a high probability of importance for clinical pharmacist requirements [20, 21]. In addition, a network funnel plot was used to evaluate publication bias in this study, and all the data in this study were processed using R 4.4.1 software and STATA 15.1 software.

Results

Section I: Cross-sectional study

Demographics of study participants

A total of 600 participants completed the questionnaire; owing to the evaluation of this questionnaire as "low-risk" by the Ethics Committee of Shenyang Pharmaceutical University, the participants' responses to the questionnaire represented their informed consent. Among them, 89 participants were clinical pharmacy undergraduates in their fourth year, and 114 participants were in their fifth year. A total of 181 participants were clinical pharmacy

Table 1 Formative survey from individuals with different identities for basic chemistry education

	Identity	Clinical Pharmacy Undergraduate—Fourth Year(n = 89)	Clinical Pharmacy Undergraduate—Fifth Year(n = 114)	Clinical Pharmacy Graduate (n = 181)	Current clinical pharmacist-On duty(n = 106)	Have experience in clinical pharmacist training(n = 60)	Other(n = 50)	P vaule inner-group
Demographics Information	Age(mean ± SD)	22.17 ± 2.11	23.29 ± 2.31	24.95 ± 2.11	34.87 ± 5.59	34.56 ± 5.78	31.15 ± 8.76	
	Gender(male/female/unknown)	32/56/1	43/69/2	61/119/1	43/62/1	28/31/1	18/32	
	Is there a disconnect between theory and practice in basic chemistry education?(Yes/No)	25/64	47/67	80/101	69/37	38/27	24/26	
	What do you think of the weight of basic chemistry education? (Moderate/Should be strengthened/Should be weakened)	64/9/16	78/19/17	127/36/18	61/21/24	43/4/13	31/12/7	0.026*
Importance of Eight Chemistry	Importance of Analytical Chemistry (Important/General/Not Important)	58/26/5	63/44/7	111/62/8	59/37/10	29/24/7	34/14/2	0.372
	Importance rate	65.17%	55.26%	61.33%	55.66%	48.33%	68.00%	
	P value (adjusted by age and gender)	0.769	0.403	0.916	0.211	0.193	-	
	Importance of Inorganic Chemistry (Important/General/Not Important)	52/30/7	38/59/17	73/89/19	32/57/17	18/33/9	26/18/6	0.004*
	Importance rate	58.43%	33.33%	40.33%	30.19%	30%	52.00%	
	P value (adjusted by age and gender)	0.403	0.032*	0.237	0.020*	0.084*	-	
	Importance of Organic Chemistry (Important/General/Not Important)	63/22/4	67/40/7	126/50/5	60/34/12	34/21/5	30/15/5	0.097
	Importance rate	70.78%#	58.77%	69.61%	56.60%	56.67%	60.00%	
	P value (adjusted by age and gender)	0.128	0.983	0.111	0.870	0.984	-	
	Importance of Medicinal Chemistry (Important/General/Not Important)	74/14/1	86/24/4	145/30/6	76/23/7	46/10/4	39/7/4	0.437
	Importance rate	83.15%#	75.44%#	80.11%#	71.70%#	76.67%#	78.00%#	
	P value (adjusted by age and gender)	0.401	0.881	0.457	0.653	0.938	-	
	Importance of Medicinal Chemistry Experiment (Important/General/Not Important)	64/23/2	63/41/10	111/55/15	46/47/13	31/19/10	24/19/7	0.003*
	Importance rate	71.91%#	55.26%	61.33%	43.40%	51.67%	48.00%	
	P value (adjusted by age and gender)	0.003*	0.236	0.040*	0.971	0.407	-	
	Importance of Physical Chemistry (Important/General/Not Important)	44/24/21	32/59/23	74/63/44	30/58/18	16/29/15	18/25/7	0.001*
	Importance rate	48.44%	28.07%	40.88%	28.30%	26.67%	36.00%	
	P value (adjusted by age and gender)	0.459	0.500	0.795	0.257	0.298	-	
	Importance of Biochemistry (Important/General/Not Important)	71/15/3	86/27/1	159/19/3	76/27/3	39/17/4	38/11/1	0.008*
	Importance rate	79.78%#	75.44%#	87.85%#	71.70%#	65.00%	76.00%#	
	P value (adjusted by age and gender)	0.428	0.705	0.009*	0.803	0.656		
	Importance of Natural Medicinal Chemistry (Important/General/Not Important)	48/29/12	42/55/17	95/68/18	46/42/18	24/27/9	21/22/7	0.247
	Importance rate	53.93%	36.84%	52.49%	43.40%	40.00%	42.00%	

Table 1 (continued)

	Identity	Clinical Pharmacy Undergraduate—Fourth Year(n = 89)	Clinical Pharmacy Undergraduate—Fifth Year(n = 114)	Clinical Pharmacy Graduate (n = 181)	Current clinical pharmacist-On duty(n = 106)	Have experience in clinical pharmacist training(n = 60)	Other(n = 50)	P vaule inner-group
	P value (adjusted by age and gender)	0.081	0.674	0.012*	0.952	0.983		
Application of Chemical Knowledge in Clinical Pharmacist Work	Have you read the structure of the instruction manual(Yes/No)	65/24	73/41	120/61	67/39	36/24	34/16	0.613
	Yes rate	73.03%#	64.04%	66.30%	63.21%	60.00%	68.00%	
	P value (adjusted by age and gender)	0.877	0.151	0.373	0.703	0.739		
	Have you considered the metabolic properties of the drug based on its structural formula(Yes/No)	66/23	76/38	113/68	63/43	42/18	33/17	0.302
	Yes rate	74.16%#	66.67%	62.43%	59.44%	70.00%#	66.00%	
	P value (adjusted by age and gender)	0.407	0.934	0.853	0.513	0.367		
	Have you consider drug solubility or lipid solubility based on the structural formula(Yes/No)	72/17	84/30	119/62	68/38	36/24	35/15	0.044*
	Yes rate	80.90%#	73.68%#	65.75%	64.15%	60.00%	70.00%	
	P value (adjusted by age and gender)	0.388	0.892	0.399	0.837	0.864		
	Have you consider drug side effects been considered based on the structural formula(Yes/No)	71/18	81/33	110/71	58/48	34/26	34/16	0.002*
	Yes rate	79.78%#	71.05%#	60.77%	54.72%	56.67%	68.00%	
	P value (adjusted by age and gender)	0.206	0.946	0.350	0.256	0.569		
	Have you ever felt that knowledge of medicinal chemistry is very important in your clinical practice as a clinical pharmacist?(Yes/No)	78/11	94/20	151/30	73/33	49/11	42/8	0.016*
	Yes rate	87.64%#	82.46%#	83.43%#	68.87%	81.67%#	84.00%#	
	P value (adjusted by age and gender)	0.594	0.625	0.756	0.690	0.268		

*significant difference; #Great impact

graduate students. There were 106 current clinical pharmacists, 60 participants who had received clinical pharmacist training, and 50 participants who had been exposed to clinical training and were currently engaged in other work. The demographic characteristics indicated that there were more female participants in every group, with an average age greater than 20 years for the student groups and greater than 30 years for the clinical pharmacist groups. Most participants in each group believed that there was no disconnect between theory and practice in basic chemistry education and that the weight of basic chemistry courses was still acceptable (Table 1).

Importance of eight types of chemistry

With respect to the importance of the eight major chemistry disciplines, at a proportion of selection importance greater than 70%, we found that medicinal chemistry and biochemistry were relatively important among the participants in all of the groups. Additionally, considering the importance of medicinal chemistry and biochemistry, we conducted an intergroup difference analysis and determined that although there were significant differences among the biochemistry groups, after excluding the influencing factors of age and gender, differences were only observed between the clinical pharmacy graduate group and other groups, which suggested that the importance of biochemistry requirements was still not disputed. In addition, fourth-year undergraduate students in clinical pharmacy also considered organic chemistry and medicinal chemistry experiment important for the requirements of clinical pharmacists. This phenomenon may have occurred because students who have just entered clinical rotation as junior clinical pharmacists may not be particularly experienced regarding the application of basic chemistry knowledge. Furthermore, the results revealed that participants felt that medicinal chemistry and biochemistry are relatively important in the requirements of clinical pharmacists, while the importance of organic chemistry and medicinal chemistry experiment are average. The respondents felt that the importance of analytical chemistry, inorganic chemistry, physical chemistry and natural medicinal chemistry is not significant (Table 1). These results preliminary indicated that for the professional needs of clinical pharmacists, knowledge of medicinal chemistry and biochemistry were more important.

Application of chemical knowledge in clinical pharmacist work

We have achieved innovative results in the application of chemistry knowledge in the work of clinical pharmacists. First, only fourth-year undergraduate students would read the structural formula of the medication in the instruction manual and consider the metabolic characteristics of the drug on the basis of its structural formula. Second, only fourth- and fifth-year undergraduate students would consider drug solubility or lipid solubility on the basis of structural formulas and consider drug side effects on the basis of structural formulas. Finally, only current clinical pharmacists rarely felt that there was a moment in their work where chemical knowledge was more important (Table 1). The above results demonstrated that basic chemistry knowledge was more important for junior clinical pharmacists.

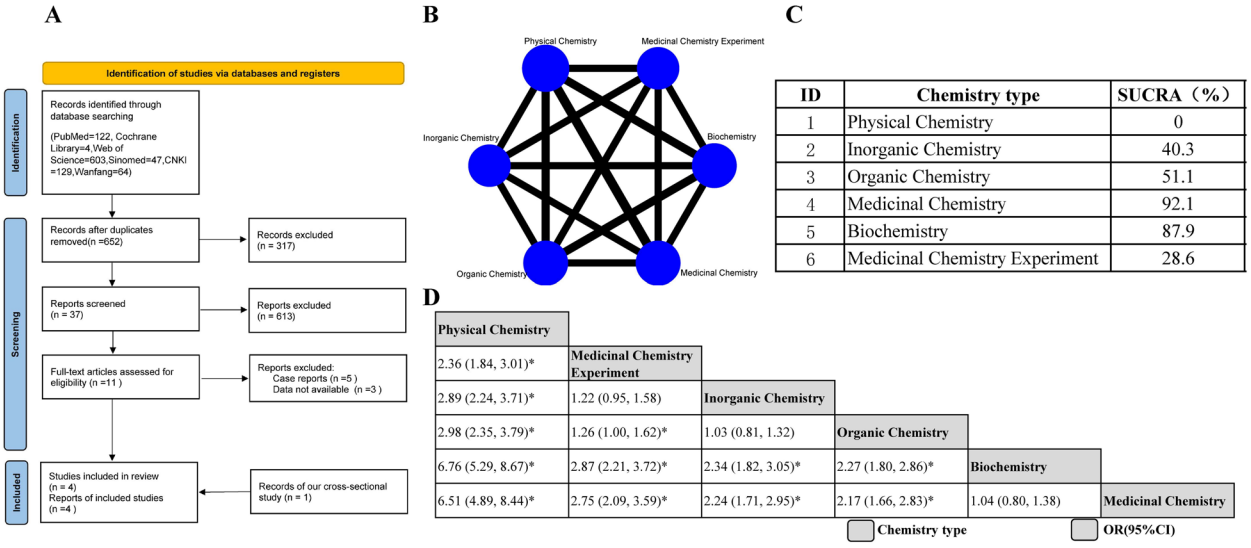
Second II: Narrative review

Importance of chemistry education outcomes

To further demonstrate the important roles of medicinal chemistry and biochemistry in clinical pharmacist requirements, we conducted a narrative review. First, after layers of literature screening, only three articles were included in this narrative review; thus, the basic chemical importance ratios of the three included studies with acceptable quality were extracted and classified [22–24] (Fig. 1A, Table 2, Table S3). In addition, we incorporated the data from our questionnaire survey to construct a network plot and rank the importance of eight types of basic chemistry education (Fig. 1B). In summary, we found that, similar to the outcomes of the questionnaire survey, medicinal chemistry ranked first at 92.1%, and biochemistry ranked second at 87.8%, followed by organic chemistry, inorganic chemistry, experimental medicinal chemistry and physical chemistry (Fig. 1C). From the league chart, we observed significant differences between medicinal chemistry and physical chemistry, indicating significant differences in the importance of different types of basic chemistry, and there were significant differences among the most closely related groups (Fig. 1D). On the basis of the above narrative review results, consistent with the questionnaire surveys, it was determined that medicinal chemistry and biochemistry play particularly important roles in the needs of clinical pharmacists.

Discussion

Our study is a mixed-methods study. Section I, through a cross-sectional survey, we demonstrated that from the perspectives of undergraduate and graduate students in clinical pharmacy, current clinical pharmacists, and those who have undergone clinical pharmacy training, basic knowledge of medicinal chemistry and biochemistry is highly important for the professional requirements of clinical pharmacists. Moreover, students majoring in clinical pharmacy as junior clinical pharmacists refer to the structural formula in the instruction manual to speculate on solubility and adverse reactions. Most of our participants believed that knowledge of chemistry is important (Table 1). Section II, we further determined



ID	Chemistry type	SUCRA（%）
1	Physical Chemistry	0
2	Inorganic Chemistry	40.3
3	Organic Chemistry	51.1
4	Medicinal Chemistry	92.1
5	Biochemistry	87.9
6	Medicinal Chemistry Experiment	28.6

Fig. 1 Chemistry education importance outcomes from narrative review. **A**Flow chart of included studies and quality assessment, **B**Network plot of various basic chemistry education types, **C** SUCRA scoring results for various basic chemistry education types, **D**Network league figure for importances in various basic chemistry education types

Table 2 Basic data for narrative review

Author, year	Chemistry type	Important	Not-important	All
Alsharif N Z, 2020 [21]	Medicinal Chemistry	90	9	99
Zhu L, 2021 [22]	Analytical Chemistry	92	7	99
	IInorganic Chemistry	132	159	291
	Organic Chemistry	135	156	291
	Medicinal Chemistry	147	144	291
	Medicinal Chemistry Experiment	192	99	291
	Physical Chemistry	159	132	291
	Biochemistry	126	165	291
	Natural Medicinal Chemistry	174	117	291
Masato Okubo, 2017 [23]	Physical Chemistry	138	153	291
	Organic Chemistry	22	420	442
	Biochemistry	44	398	442

the importance of medicinal chemistry and biochemistry for the professional demands of clinical pharmacists through narrative review (Fig. 1). These two chemistry courses, when combined with pharmacy and medical courses, can become characteristic courses in clinical pharmacy.

Our research revealed that basic knowledge of medicinal chemistry and biochemistry are most important for the professional needs of clinical pharmacists. We refer to two cases that are highly likely to be encountered in clinical rotation. The first case involves a coronary heart disease patient who underwent coronary stent implantation surgery and had underlying gastrointestinal diseases, resulting in decreased gastric acid secretion. The doctor planned to use clopidogrel to prevent thrombosis. From the perspective of drug solubility and medicinal chemistry (structural chemistry), clopidogrel molecules contain ester groups that are activated into effective metabolites through liver enzyme metabolism. Due to the reduced gastric acid secretion and increased gastric pH in the patient, the solubility of clopidogrel raw material may be reduced, affecting its absorption and thus affecting the generation of active metabolites [25, 26]. Clinical pharmacists remind physicians to give attention to the possibility of inadequate drug absorption due to insufficient gastric acid secretion in patients on the basis of the chemical properties of clopidogrel and to moderately increase the dose of clopidogrel on the basis of platelet monitoring results. The second case involves a patient with active tuberculosis and liver dysfunction. It is necessary to use rifampicin for anti-tuberculosis treatment. From the perspective of drug structure, rifampicin is a macrolide antibiotic that contains hydroxyl and ester groups and has strong lipid solubility. It is widely metabolized by the liver and induces the activity of the liver cytochrome P450 enzyme system, which may accelerate its own metabolism and those of other drugs. Therefore, in patients with liver function impairment, the metabolic capacity of the liver decreases, and the risk

of accumulation of rifampicin and its metabolites in the body increases, which may cause toxic reactions [27, 28]. The above content proves the importance of basic knowledge of pharmaceutical chemistry for the professional needs of clinical pharmacists. Moreover, biochemical knowledge is related to protein function, for example, the shape of enzyme catalytic pockets and substrate binding specificity that directly determine the mechanism of action of drug molecules. Obtaining high-precision protein structures through X-ray crystallography and cryo-electron microscopy can reveal key amino acid residues at drug binding sites, providing clear targets for drug design [29]. In summary, knowledge of medicinal chemistry and biochemistry is crucial for the requirements of clinical pharmacists.

Our teaching team plans to adopt a problem-based learning (PBL) teaching method in the future, guiding the integration of medicinal chemistry with pharmacology, pharmacy, and clinical medicine to improve students' comprehensive analytical abilities. For example, we can ask students questions similar to the following: The nonselective β blocker propranolol increases the risk of portal vein thrombosis, whereas the selective β1 blocker metoprolol has more advantages in reducing heart rate and blood pressure. Do you know why this phenomenon occurs? Is it because of different structures [30]? Why is polyene phosphatidylcholine only compatible with a glucose solvent in clinical applications? Is it also determined by knowledge of medicinal chemistry [31]? Through PBL teaching, clinical pharmacy majors can not only master the theoretical knowledge of medicinal chemistry but also apply it to clinical pharmacy practice, enhancing their ability to solve complex medication problems.

With respect to the application of chemical knowledge by clinical pharmacists, we found that only undergraduate and graduate students majoring in clinical pharmacy would examine the structural formula and continue to consider issues such as solubility and adverse reactions based on the structural formula, whereas senior in-service clinical pharmacists consider that chemical knowledge as not very important. These results were unexpected, and there is no other supporting evidence in publications. We speculate that this is because undergraduate and graduate students who enter clinical rotation to participate in clinical pharmacist work are more likely to apply their knowledge of medicinal chemistry to their clinical pharmacist work and are more likely to combine their learned knowledge to infer solubility and adverse reactions from chemical structural formulas. However, current clinical pharmacists have a thorough understanding of issues such as lipid solubility, water solubility, and adverse reactions, and the above knowledge does not require association through chemical structural formulas. Therefore, we speculate that junior clinical pharmacists, including students, are more likely to rely on chemical knowledge when they first start as clinical pharmacists, but as they grow into mature clinical pharmacists, the importance of their basic chemical knowledge decreases.

This mixed-methods study has several limitations. First, from the perspective of a cross-sectional study, although our sample size was relatively large, our questionnaire survey focused on students and graduates from a pharmaceutical university in Northeast China, and we are unable to conduct surveys with clinical pharmacists and students throughout China or relevant participants from other countries. This is a geographical limitation of our research. Second, the participants included in our questionnaire survey were generally young; the clinical pharmacist group was the oldest, with an average age of only 34.87 years. This is because clinical pharmacists belong to an emerging industry in China, and there are fewer elderly clinical pharmacists. Therefore, they may have a good understanding and memory of basic knowledge, making it easier for them to remember drug properties and adverse reactions. This is why established clinical pharmacists generally no longer consider chemical factors. Third, there are relatively few published articles on the importance of various chemistry disciplines in meeting the needs of clinical pharmacists, which may be because it is a research direction that has been less considered. Therefore, only four articles were included in our narrative review. Finally, as clinical pharmacists, there may be some participants with advanced qualifications, which means that they are older and have received relatively superficial basic chemistry education, which may also be a limitation of our research. We look forward to more high-level research on the importance of various chemistry subjects in the future, which can support our research results.

Conclusions

In conclusion, our research confirms for the first time that medicinal chemistry and biochemistry knowledge are particularly important among all basic chemistry disciplines for the requirements of clinical pharmacists, and the mastery of this knowledge is even more important for the application demands of junior clinical pharmacists. Future research is expected to increase the teaching hours of medicinal chemistry and biochemistry and the integration of these subjects more organically with the basic disciplines of pharmacy and medicine to create a distinctive educational system for medicinal chemistry and biochemistry in clinical pharmacy that better meets the professional needs of clinical pharmacists.

Abbreviations

AHRQ American Agency for Healthcare Research and Quality
CI Confidence interval

CNKI	China National Knowledge Infrastructure
OR	Odds ratio
PBL	Problem-based learning
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-analysis
SUCRA	Surface under the cumulative ranking curve

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12909-025-08555-z.

Supplementary Material 1. Table S1 Questionnaire.

Supplementary Material 2. Table S2 Search Strategy.

Supplementary Material 3. Table S3 American Agency for Healthcare Research and Quality (AHRQ) scale.

Supplementary Material 4. Fig.S1 Network funnel plot.

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None.

Authors' contributions
Conceptualization, methodology, investigation, data Curation, formal analysis, writing – Original draft, writing – review & editing, supervision: ZYS, XZA, ZMY and ZFY. Methodology, investigation, writing-review & editing: TZM, ZYK and XXB. Data curation, formal analysis, writing – original draft: SXH. All authors read and approved the final manuscript.

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Data availability
All data generated or analysed during this study are included in this published article [and its supplementary information files].

Declarations

Ethics approval and consent to participate
The research plan and questionnaire design in this study have been submitted to the Ethics Committee of Shenyang Pharmaceutical University. The Ethics Committee of Shenyang Pharmaceutical University considered that this cross-sectional study only involves teaching related knowledge content and does not involve research on human health information, psychology, and social behavior, nor does it involve the personal privacy and commercial interests of the subjects. Therefore, according to the "Ethical Review Measures for Biomedical Research Involving Human Subjects", this study waives under the category of exempt subjects from informed consent.
This research adheres to the relevant principles of the Helsinki Declaration, respects the rights and dignity of participants, ensures that all participants participate voluntarily with full knowledge, and protects their personal privacy and data security.

Consent for publication
Not applicable.

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The authors declare no competing interests.

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